TABLE I

CSO CONTACT GUIDANCE: SEVEN OBJECTS, 5 PAIRED TRIALS PER OBJECT AND CONFIGURATION. TIMES ARE MEAN ± POPULATION STD; EXECUTION TIME IS SUCCESS-ONLY WALL TIME, AND PLANNING TIME COVERS ALL ROUNDS.

Metric	Contact Model		Solvers			**Ours (SCM+SQP)**
	MuJoCo QP	LCP	NLP (IPOPT)	CMA-ES	CEM	
Success Rate	40.0%	20.0%	42.9%	42.9%	42.9%	45.7%
Execution Time (s)	138.75 ± 165.29	46.51 ± 26.84	532.48 ± 505.12	521.84 ± 394.56	312.30 ± 237.63	19.14 ± 20.43
Planning Time (ms)	123.45 ± 39.15	117.03 ± 29.94	500.12 ± 203.16	734.27 ± 174.99	442.67 ± 102.99	29.52 ± 10.94